

Onatola Timilehin Faruq

Blockchain Integrations Engineer | Cross-Chain Infrastructure | ZK Developer

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Executive Summary

Blockchain engineer specializing in cross-chain infrastructure, bridge protocols, and interoperability solutions. Built OmniSwap SDK enabling atomic swaps across 6 heterogeneous blockchains and Cybria Bridge with multi-sig validator architecture. Production Solidity experience on Ethereum, Arbitrum, and Base with intermediate Rust proficiency from Solana/Anchor development. ZK experience building Nocturne using zero-knowledge proofs and stealth addresses. Ships complete integrations independently with comprehensive documentation and test coverage.

Core Competencies: Cross-Chain Integration • Bridge Protocols • Zero-Knowledge Systems • Smart Contracts (Solidity/Rust) • Rollup Development • Multi-Chain Architecture

Technical Expertise

Smart Contracts: Solidity (Advanced) • Rust/Anchor (Intermediate) • Noir • Clarity

L1/L2 Networks: Ethereum • Arbitrum • Base • Optimism • Solana • Hedera • Cosmos/Osmosis • Mina • Stacks

Cross-Chain: HTLC Atomic Swaps • Bridge Protocols • Multi-sig Validators • Cross-Chain Messaging • Chain Adapters

Zero-Knowledge: zkSNARKs • zkSTARKs • Noir Circuits • Stealth Addresses (DKSAP) • Privacy Pools

DeFi Protocols: Uniswap v3/v4 • Aave v3 • Curve • GMX • Chainlink (Oracles, CCIP)

Infrastructure: Foundry • Hardhat • IPFS • Docker • GitHub Actions • CI/CD

Cross-Chain & Integration Projects

OmniSwap SDK

Cross-Chain Atomic Swap SDK | TypeScript/Solidity | github.com/Leihyn/omniswap-sdk

Production-grade SDK enabling atomic swaps across 6 heterogeneous blockchains.

- Architected HTLC-based swap protocol with automated refund management and failure recovery
- Built chain-specific adapters for Zcash, Osmosis, Mina, Aztec, Miden, Fhenix
- Implemented unified interface abstracting chain differences for seamless integration
- Created robust error handling with typed errors and exponential backoff retry logic
- Authored 70KB+ documentation with architecture overview, API reference, and 5 working examples
- Set up CI/CD pipeline with GitHub Actions for automated testing across chains

Cybria Bridge

Cross-Chain Bridge | EVM/Python | github.com/Leihyn/cybria-bridge

Cross-chain bridge with multi-signature security model.

- Designed 3-of-5 multi-sig validator architecture for secure cross-chain transfers
- Implemented message passing protocol with documented trust assumptions and failure modes
- Built validator coordination system for cross-chain state verification

OmniShield Wallet

Multi-Chain HD Wallet | TypeScript | github.com/Leihyn/omnishield-wallet

HD wallet supporting 6 blockchains with unified transaction interface.

- Implemented BIP-39/32/44 key derivation across heterogeneous chain types
 - Built chain-agnostic adapter interface for balance queries and transactions
 - Integrated with OmniSwap SDK for cross-chain swap functionality
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Zero-Knowledge Projects

Nocturne

Private Payments | Noir/Solana | github.com/Leihyn/nocturne

Private payment system using zero-knowledge proofs on Solana.

- Implemented stealth addresses using DKSAP (Dual-Key Stealth Address Protocol)
- Built ZK circuits for transaction privacy with Noir
- Designed privacy pool architecture with timing decorrelation

Sentiment – Uniswap v4 Hook

UHI7 Incubator | Solidity | github.com/Leihyn/Sentiment

Dynamic fee hook for Uniswap v4 with protocol-level integration.

- Graduated from UHI7 (15% acceptance rate) with Uniswap Labs collaboration
 - Implemented hook integration with Uniswap v4 core contracts
 - Built multi-keeper architecture using Chainlink Automation + Gelato
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Professional Experience

Blockchain Engineer | DeFiConnectCredit | Remote | Jan 2025 – Present

- Architected modular smart contract systems integrating Aave v3, Uniswap v3, GMX, and Curve on Arbitrum and Base
- Implemented cross-chain asset management reducing bridging latency by 40%
- Established Foundry testing framework with 95%+ coverage
- Reduced transaction costs by 35% through assembly optimization

Blockchain Development Engineer (Intern) | DeFiConnectCredit | Remote | Jun 2024 – Dec 2024

- Contributed to core lending protocol with collateral management and liquidation mechanics
 - Conducted security reviews on 15+ contracts identifying cross-contract vulnerabilities
 - Deployed ERC721 system with Merkle tree proofs for 500+ users
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Additional Projects

Project	Tech	Description
Comic Pad	Hedera/React	Solo-built NFT marketplace with HTS token integration
TruthBounty	BNB/Solidity	Prediction market reputation protocol (2nd place Seedify)
USDCx-Lending	Stacks/Clarity	Lending protocol on non-EVM chain
BankOfCronos	Cronos/Solidity	DeFi protocol with automated strategies
CelestiFi	TypeScript	Whale tracking with cross-chain monitoring

Achievements & Certifications

2nd Place – Seedify Prediction Market Hackathon

Uniswap Hook Incubator (UHI7) – Graduate 15% acceptance rate • Protocol-level integration • Uniswap Labs collaboration

Cyfrin Updraft – DeFi Security & Protocol Engineering Aave v3 • Uniswap v3/v4 • GMX • Curve • Smart Contract Security

Hashgraph Developer Certification – The Hashgraph Association

Education

Bachelor of Science in Physiotherapy University of Ibadan, College of Medicine | Graduated 2024

Languages

Human: English (Native) • Spanish (Conversational)

Programming: Solidity • TypeScript • Rust • Clarity • Noir

Open Source

- **11 public repositories** with comprehensive documentation and CI/CD
 - All integration projects include architecture docs, API references, and working examples
 - Comfortable working in public with community feedback and contributions
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Portfolio: faruukku.vercel.app / github.com/Leihyn / References available upon request